

² For customization of ESP32-S3-WROOM-1-H4, ESP32-S3-WROOM-1U-H4, and ESP32-S3-WROOM-1U-N16R16VA, please [contact us](#).

³ By default, the SPI flash on the module operates at a maximum clock frequency of 80 MHz and does not support the auto suspend feature. If you have a requirement for a higher flash clock frequency of 120 MHz or if you need the flash auto suspend feature, please [contact us](#).

⁴ The modules use PSRAM integrated in the chip's package. For specifications, refer to Section [6.5 Memory Specifications](#).

⁵ Ambient temperature specifies the recommended temperature range of the environment immediately outside the Espressif module.

⁶ For details, refer to Section [10.1 Module Dimensions](#).

⁷ Please note that the VDD_SPI voltage is 1.8 V for ESP32-S3-WROOM-1-N16R16VA and ESP32-S3-WROOM-1U-N16R16VA only.

At the core of the modules is an ESP32-S3 series of SoC, an Xtensa® 32-bit LX7 CPU that operates at up to 240 MHz. You can power off the CPU and make use of the low-power co-processor to constantly monitor the peripherals for changes or crossing of thresholds.

Note:

For more information on ESP32-S3, please refer to [ESP32-S3 Series Datasheet](#).

For chip revision identification, ESP-IDF release that supports a specific chip revision, and other information on chip revisions, please refer to [ESP32-S3 Series SoC Errata](#) > Section *Chip Revision Identification*.

1.3 Applications

- Smart Home
- Industrial Automation
- Health Care
- Consumer Electronics
- Smart Agriculture
- POS Machines
- Service Robot
- Audio Devices
- Generic Low-power IoT Sensor Hubs
- Generic Low-power IoT Data Loggers
- Cameras for Video Streaming
- USB Devices
- Speech Recognition
- Image Recognition
- Wi-Fi + Bluetooth Networking Card
- Touch and Proximity Sensing

6.3 SPI interface

The SPI interface is compatible with SPI mode ‘00’ (CPOL = CPHA = ‘0’) and mode ‘11’ (CPOL = CPHA = ‘1’). The automatic selection between mode ‘00’ and ‘11’ is determined by the value of SCK after the CSB falling edge.

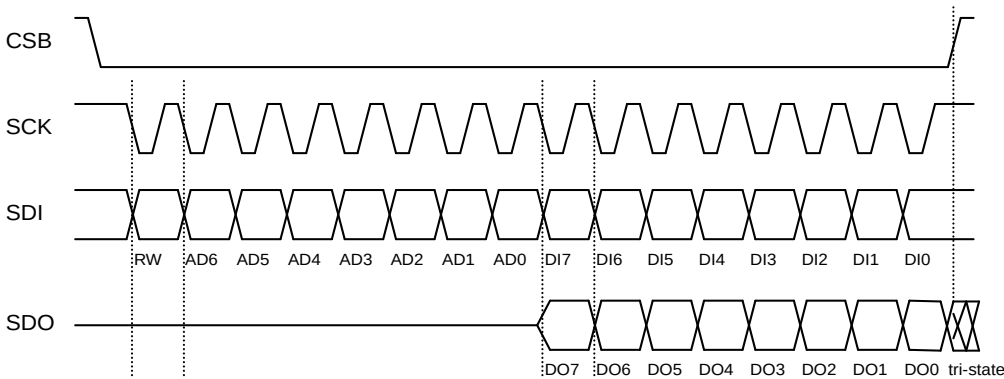
The SPI interface has two modes, namely 4-wire and 3-wire mode. However, the protocol is the same for both. The 3-wire mode is selected by setting ‘1’ to the register spi3w_en. The pad SDI is used as a data pad in 3-wire mode.

The SPI interface uses the following pins:

- CSB: chip select, active low
- SCK: serial clock
- SDI: serial data input; data input/output in 3-wire mode
- SDO: serial data output; hi-Z in 3-wire mode

For more connection instructions, please refer to Chapter 7.

CSB is active low and has an integrated pull-up resistor. Data on SDI is latched by the device at SCK rising edge and SDO is changed at SCK falling edge. Communication starts when CSB goes to low and stops when CSB goes to high; during these transitions on CSB, SCK must be stable. The SPI protocol is shown in It is important to note that Picture 6. For timing details, please review Table 23.



Picture 6: SPI protocol (shown for mode ‘11’ in 4-wire configuration)

It is important to note that in the SPI mode, only 7 bits of the register addresses are used; the MSB of register address is not used and replaced by a read/write bit (RW = ‘0’ for write and RW = ‘1’ for read). For example, address 0xF7 is accessed by using SPI register address 0x77. On the one hand, the byte 0x77 is transferred for write access, and on the other hand, the byte 0xF7 is transferred for read access.

6.3.1 SPI write

Writing is done by lowering CSB and sending pairs control bytes and register data. The control bytes consist of the SPI register address (= full register address without bit 7) and the write command (bit7 = RW = ‘0’). Several pairs can be written without raising CSB. The transaction is ended by a raising CSB. The SPI write protocol is depicted in Picture 7.

		Control byte							Data byte									Control byte							Data byte								
Start	RW	Register address (F4h)							Data register - address F4h							RW		Register address (F5h)							Data register - address F5h							Stop	
CSB = 0	0	1	1	1	0	1	0	0	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	0	1	1	1	0	1	0	1	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	CSB = 1

Picture 7: SPI multiple byte write (not auto-incremented)

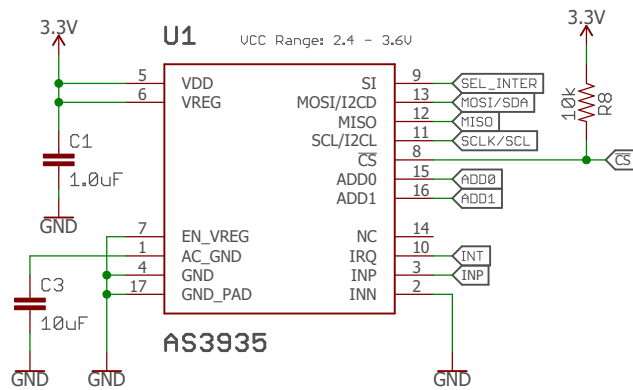
Technical Index

Parameter	Index	unit
Range of measurement	0.3~1.0; 1.0~2.5; 2.5~10	Micrometer ($\mu\text{ m}$)
Counting Efficiency	50%@0.3 $\mu\text{ m}$ 98%@ $\geq 0.5\mu\text{ m}$	
Effective Range (PM2.5 standard)	0~500	$\mu\text{ g/m}^3$
Maximum Range (PM2.5 standard) *	≥ 1000	$\mu\text{ g/m}^3$
Resolution	1	$\mu\text{ g/m}^3$
Maximum Consistency Error (PM2.5 standard data)*	$\pm 10\% @ 100 \sim 500 \mu\text{ g/m}^3$ $\pm 10 \mu\text{ g/m}^3 @ 0 \sim 100 \mu\text{ g/m}^3$	
Standard Volume	0.1	Litre (L)
Single Response Time	< 1	Second (s)
Total Response Time	≤ 10	Second (s)
DC Power Supply	Typ:5.0 Min:4.5 Max: 5.5	Volt (V)
Active Current	≤ 100	Milliampere (mA)
Standby Current	≤ 200	Microampere ($\mu\text{ A}$)
Interface Level	L $< 0.8 @ 3.3$ H $> 2.7 @ 3.3$	Volt (V)
Working Temperature Range	-10~+60	$^{\circ}\text{C}$
Working Humidity Range	0~99%	
Storage Temperature Range	-40~+80	$^{\circ}\text{C}$
MTTF	≥ 3	Year (Y)
Physical Size	50×38×21	Millimeter (mm)

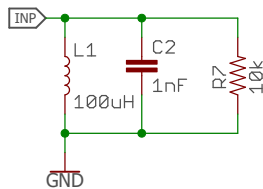
Note 1: Maximum range means that the highest output value of the PM2.5 standard data is not less than 1000.

Note 2: "PM2.5 standard data" is the "data2" in the appendix.

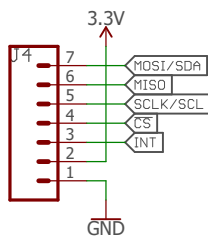
AS3935 Lightning Detector



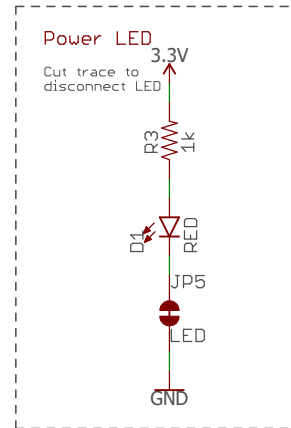
500 kHz Resonance Antenna



Header

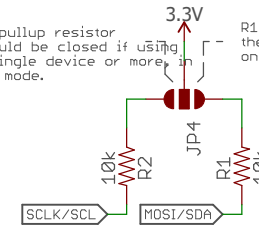


Jumpers



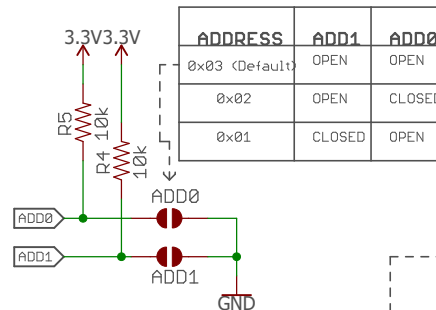
I2C Pull Up Resistors

R2 pullup resistor should be closed if using a single device or more in I2C mode. R1 should only be closed when there are three or more devices on the I2C bus



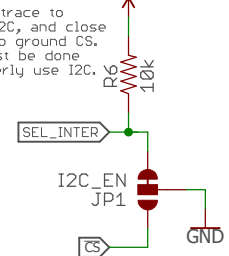
Address Jumpers

Close jumpers to change I2C Address



Select Interface

Cut top trace to select I2C, and close bottom to ground CS. Both must be done to properly use I2C.



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TITLE: SparkFun_AS3935_Lightning_Detector

Design by:
Elias Santistevan

REV:
v20

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Sheet: 1/1